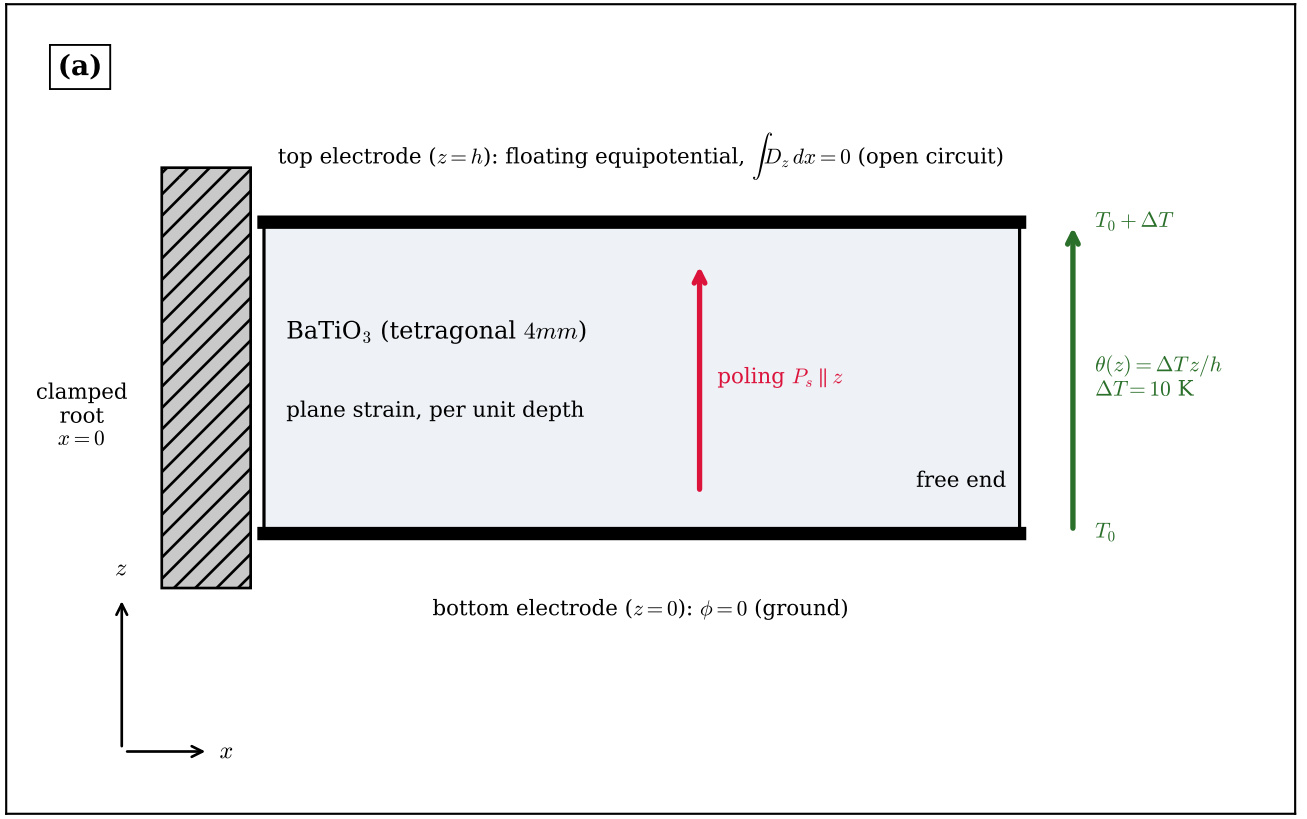
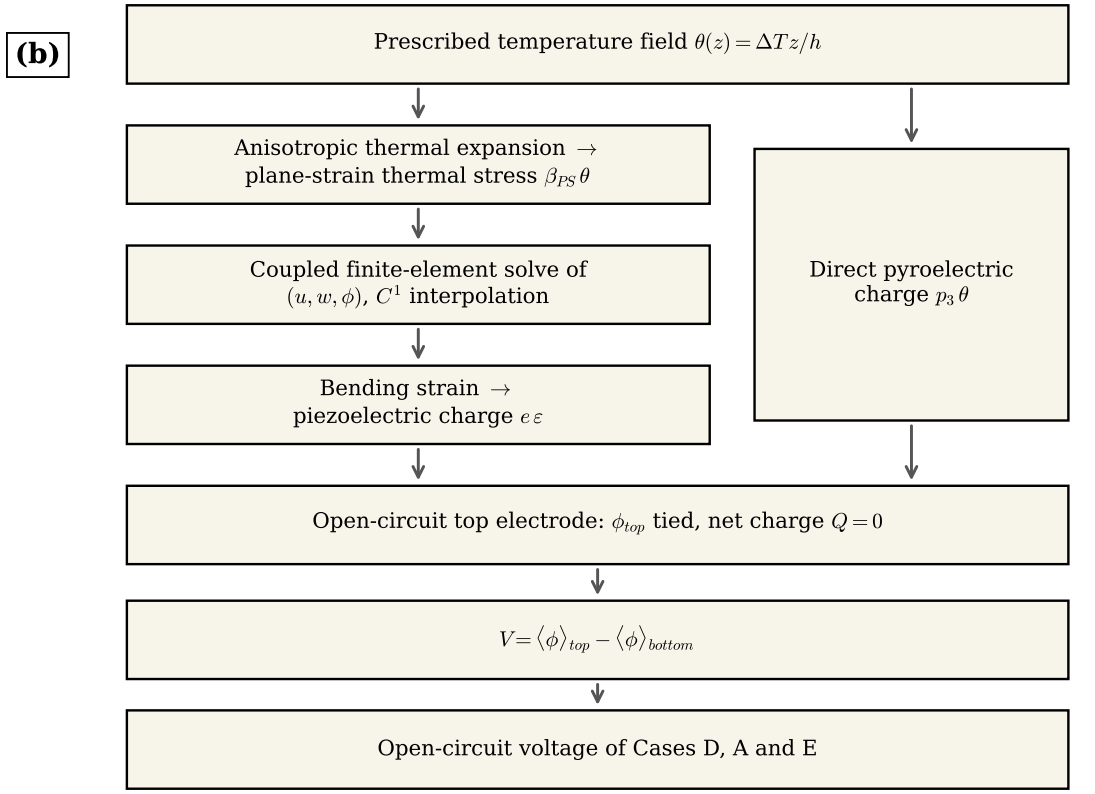

(a)
top electrode ($z=h$): floating equipotential, $\int D_z\,dx=0$ (open circuit)
$T_0+\Delta T$
$\theta(z)=\Delta T z/h$
$\Delta T=10$ K
$BaTiO_3$ (tetragonal $4mm$)
poling $P_s \parallel z$
plane strain, per unit depth
free end
clamped
root
$x=0$
T_0
z
x
bottom electrode ($z=0$): $\phi=0$ (ground)


(b)
Prescribed temperature field $\theta(z)=\Delta T z/h$
Anisotropic thermal expansion $\rightarrow$
plane-strain thermal stress $\beta_{PS}\theta$
Coupled finite-element solve of
(u, w, ϕ), C^1 interpolation
Bending strain $\rightarrow$
piezoelectric charge $e\,\varepsilon$
Direct pyroelectric
charge $p_3\,\theta$
Open-circuit top electrode: ϕ_{top} tied, net charge $Q=0$
$V=\langle\phi\rangle_{top}-\langle\phi\rangle_{bottom}$
Open-circuit voltage of Cases D, A and E